

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Trajectory Optimization

Abstract—Prior work introduced the occlusion-aware target-tracking and obstacle-avoidance problem for quadrotors and solved it in the continuous model-predictive-control (MPC) domain. We contribute a discrete reformulation of that problem class as a Higher-Order Unconstrained Binary Optimization (HUBO) problem, suitable for quantum annealing and (after quadratization) QAOA, together with a *feasibility-aware benchmarking methodology* that decomposes hard constraint satisfaction from soft trajectory quality, uses independent constraint verification, matches sampling budgets, and characterizes penalty-weight exploitation. On the 8×8 instance (960 variables, 87,684 monomials, 3,172 cubic), we benchmark classical A*, simulated annealing (SA), D-Wave neal, and Qiskit QAOA. A* and SA achieve comparable soft cost (317 vs. 303) with A* in 10 ms and SA in 47.8 s; neal is fast (0.26 s) but reaches only 22 % true feasibility, with an exploitation curve from 0 % at $\lambda=50$ to 60 % at $\lambda=2000$. QAOA reproduces the optimum on a tiny instance, included for correctness validation only. The classical baseline grounds the HUBO formulation against a conventional planner.

Index Terms—HUBO, quantum annealing, QAOA, occlusion-aware tracking, UAV trajectory, A*, feasibility, benchmarking.

I. INTRODUCTION

Visibility-aware UAV target tracking — keeping a moving target in view while avoiding obstacles — has an established literature in robotics and control. Chen *et al.* [1] plan collision-free quadrotor trajectories tracking a target in cluttered environments. Occlusion and perception have been treated explicitly: Hausman *et al.* [2] address occlusion-aware multi-robot tracking, Penin *et al.* [3] combine target tracking with collision and occlusion avoidance, Falanga *et al.* [4] fold perception objectives into MPC, and Wang *et al.* [5] extend to multi-UAV visibility-constrained tracking. Masnavi *et al.* [6] introduced the occlusion-free target-tracking and obstacle-avoidance problem solved here as a real-time multi-convex MPC. All of these are *continuous* control formulations.

In the quantum-optimization setting, QUAV [7] addresses single-UAV obstacle-avoidance path planning with QAOA on small instances, and Q4DR [8] solves drone *routing* via a hybrid QAOA-clustering + D-Wave decomposition, closer to logistics than to local motion. Neither addresses visibility-constrained trajectory planning. While prior work addressed occlusion-aware tracking in continuous MPC form, and quantum UAV papers addressed obstacle-only planning or routing, our contribution is a discrete HUBO reformulation and feasibility-aware quantum-classical benchmark for visibility-constrained UAV trajectory optimization: (1) a discrete HUBO

reformulation of the occlusion-aware target-tracking class [6], bridging continuous MPC and higher-order binary optimization; (2) a feasibility-aware benchmarking methodology with a classical A* baseline; and (3) an analysis of why the sustained-proximity term requires auxiliary variables for quadratization.

II. HUBO REFORMULATION

Let V be the grid cells and T the horizon. The decision $x[t, v] \in \{0, 1\}$ is 1 iff the UAV occupies cell v at time t . The Hamiltonian has seven terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \quad (1)$$

H_{uniq} , H_{start} , H_{move} enforce kinematic constraints quadratically; H_{obs} forbids obstacle cells.

Occlusion penalty. For each cell u we trace the Bresenham line-of-sight to the target and count intervening obstacles:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_{u \in V} x[t, u] \mid \text{LOS}(u, \text{target}) \cap O \mid, \quad (2)$$

with O the obstacle set (Fig. 1a). For a static target this is linear; for a *moving* target it becomes a joint product of UAV and target occupancy — intrinsically higher-order.

Genuinely-cubic proximity. The sustained-proximity term penalizes lingering in a buffer cell v_b for three consecutive steps:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b \in B} \sum_{t=1}^{T-1} x[t-1, v_b] x[t, v_b] x[t+1, v_b]. \quad (3)$$

Eq. (3) requires auxiliary variables for quadratization: Rosenberg reduction introduces one ancilla per cubic term [9], adding $\sim 3,172$ ancillas ($\sim 30\%$ more variables) with dense couplings that degrade the sparsity annealers exploit. The full instance has **960 variables, 87,684 monomials** (3,172 cubic).

III. FEASIBILITY-AWARE BENCHMARKING

Raw HUBO energy misleads because penalty-based solvers can lower energy by *violating* soft constraints. Our methodology adopts four safeguards: (i) energy decomposition into hard penalty and soft cost; (ii) independent constraint verification on each decoded trajectory (not energy-threshold heuristics); (iii) matched sampling budgets across stochastic solvers; and (iv) a penalty-weight sweep revealing the exploitation regime a single λ would hide. We add a **classical A* baseline** to ground the HUBO against a conventional planner.

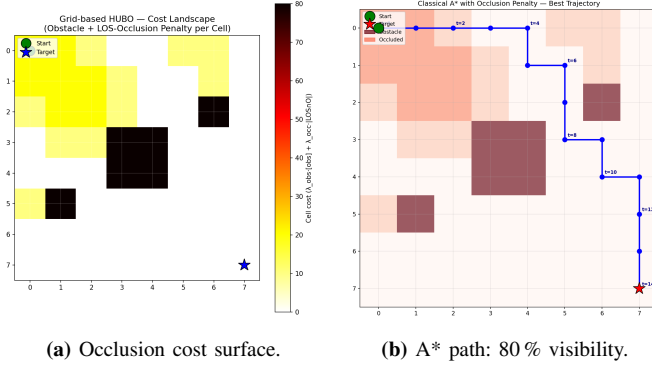


Fig. 1: (a) Per-cell obstacle + line-of-sight occlusion cost to the target; yellow cells are occluded. (b) A* trajectory: feasible, target-reaching.

TABLE I: Tier 1 head-to-head, 8×8 instance (960 variables).

Metric	A*	SA	neal
Best soft cost	317	303	496
Feasibility rate	100 %	100 %	22 %
Reaches target	yes	14 % runs	8 % runs
Visibility (% steps)	80 %	—	—
Runtime per sample	10 ms	47.8 s	0.26 s

Since Qiskit’s state-vector simulator scales as 2^n and cannot run at $n=960$, we use two tiers: **Tier 1** (full 8×8 , 960 vars: A*, SA, Neal) and **Tier 2** (tiny 2×2 , 8 vars: all solvers including QAOA). We use 50 matched samples per stochastic solver, SA at 500 sweeps, and hardened weights $\lambda=1000$. A* runs on the same grid with cell cost $1.0 + \lambda_{\text{occ}}|\text{LOS} \cap O| + \lambda_{\text{prox}}\mathbf{1}[v_b \in B]$ and Manhattan heuristic; it is an *approximate classical analogue* of the HUBO surrogate, not a term-for-term match, encoding goal-reaching through search termination rather than the explicit H_{term} cost.

IV. RESULTS

A. Tier 1 — Three Solvers, One Instance

Table I reports the matched comparison. A* and SA achieve comparable soft cost (317 vs. 303), but A* completes in 10 ms while SA needs 47.8 s — a $\sim 4,700\times$ speedup with guaranteed feasibility. Neal is fast per sample (0.26 s) but only 22 % of its samples are truly feasible, and its best feasible soft cost (496) is 56 % worse than A*’s (Fig. 1b). SA’s *mean* soft cost (~ 495) matches Neal’s *best* (496): Neal needs many reads to occasionally equal SA’s typical quality. Visibility is reported for the A* path; the stochastic benchmark logged per-sample energy and feasibility but not full traces, so an SA/Neal visibility score would require re-running with trajectory logging.

B. Constraint-Exploitation Curve

Sweeping λ (Fig. 2) reveals the failure mode the methodology targets. At $\lambda \leq 300$ Neal achieves 0 % feasibility; it rises to 13 % at $\lambda=600$, 20 % at $\lambda=1000$, and 60 % at $\lambda=2000$. A single- λ report would mis-state Neal’s capability.

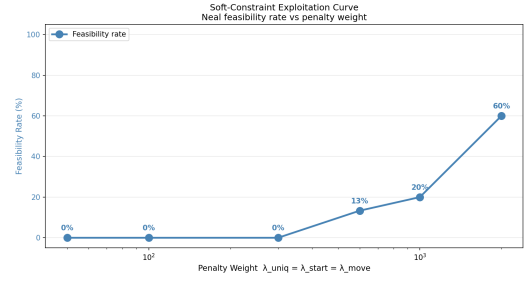


Fig. 2: Constraint-exploitation curve: Neal feasibility vs. penalty weight λ . Annealers need weights an order of magnitude above the objective scale to respect one-hot HUBO structure.

C. Tier 2 — Three-Way Comparison with QAOA

On the 8-variable instance, A*, SA, Neal, and Qiskit QAOA all reach the same optimum $E^* = -118.17$ (QuadraticProgram $\rightarrow$ QAOA, SPSA, reps=1). This tiny instance is included for *correctness validation*, not performance competition. QAOA takes 55 s vs. 0.02 s (Neal) and 0.14 s (SA): a $2,750\times$ state-vector overhead that grows exponentially with qubit count.

V. DISCUSSION AND CONCLUSION

The A* baseline answers the obvious reviewer question: the HUBO captures sensible planning structure — A* finds a feasible trajectory in milliseconds with soft cost within 5 % of the best stochastic solver. The HUBO framing’s value is not raw performance on this static-target instance; it is compatibility with quantum-style solvers and extensibility to a *moving* target, where Eq. (2) becomes intrinsically cubic and outside A*’s shortest-path framing. Feasibility-aware benchmarking proves necessary: 78 % of Neal’s hardened-weight samples violate constraints. Future work: moving target, real D-Wave hardware, and comparison with the multi-convex MPC of [6].

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